

SOP-TANK-4700: Storage Tank Entry and Cleaning

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Section 1. Purpose and Scope

Governs entry into and cleaning of storage tanks that have contained flammable, toxic, or oxygen-displacing material — a specific application of confined space entry (SOP-CSE-200) with tank-specific controls.

Section 2. Pre-Entry Isolation and Degassing

Clause 2.1 — All process connections to the tank shall be positively isolated per SOP-ISO-4500 before degassing begins; a tank shall be freed of residual product and inerted/purged to below 10% of the lower explosive limit (LEL) before hydrocarbon vapor space is opened for atmospheric testing.

Clause 2.2 — Degassing/purging progress is verified by combustible gas and toxic gas monitoring at the tank vent, not assumed complete after a fixed purge duration alone.

Section 3. Atmospheric Testing

Clause 3.1 — Full confined space atmospheric testing per SOP-CSE-200 Section 2 is required before entry, plus continuous monitoring throughout entry given the potential for residual sludge/sediment to off-gas during cleaning disturbance.

Section 4. Sludge and Residue Handling

Clause 4.1 — Tank bottom sludge is presumed hazardous waste unless characterization sampling proves otherwise; it is handled per SOP-WASTE-3900 pending characterization results.

Section 5. Static Electricity Control

Clause 5.1 — Bonding and grounding of the tank, cleaning equipment, and any non-metallic cleaning tools is required throughout cleaning of a tank that contained flammable product, given the ignition risk from residual vapor and static discharge during cleaning activity.

Section 6. Records

Clause 6.1 — Degassing verification, atmospheric monitoring logs, and waste characterization results are filed to the Mechanical Integrity File and retained for 5 years.